

Covalent Bonding

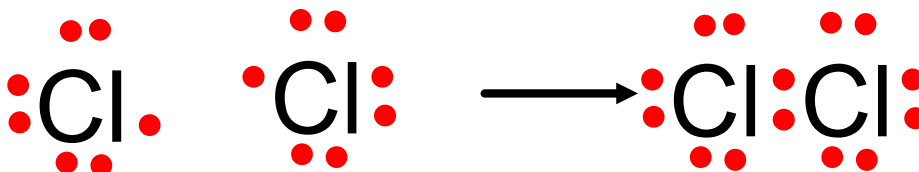
Recall: Ionic bonds are formed by metals donating electrons to a non-metal.

G.N. Lewis reasoned that an atom might attain a noble gas electron configuration by sharing electrons. A chemical bond formed by sharing a pair of electrons is called a **covalent bond**.

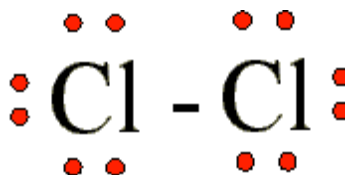
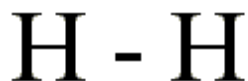
Ex - Molecular Hydrogen (H_2)



Ex - Molecular Chlorine (Cl_2)



In Lewis structures the bonding pair of electrons is usually displayed as a line, and the unshared electrons as dots.



A **molecule** is defined as a particle in which atoms are joined by covalent bonds. H_2 and Cl_2 are examples of **diatomic molecules**. Diatomic molecules consist of only two atoms.

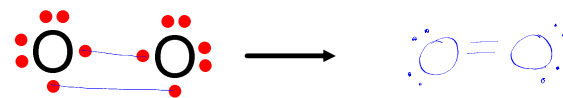


Multiple Bonds

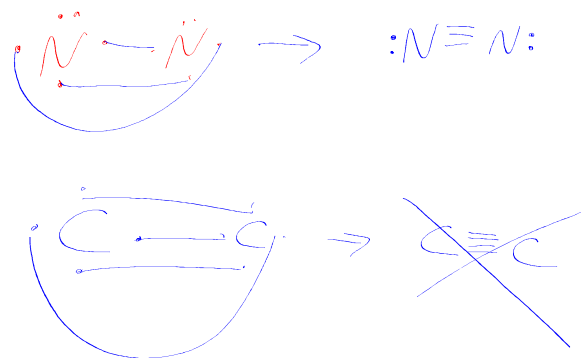
The sharing of a pair of electrons represents a single covalent bond, usually just referred to as a single bond. In many molecules atoms attain complete octets by sharing more than one pair of electrons between them.

- Two electron pairs shared a double bond
- Three electron pairs shared a triple bond

Ex - Molecular Oxygen (O_2)



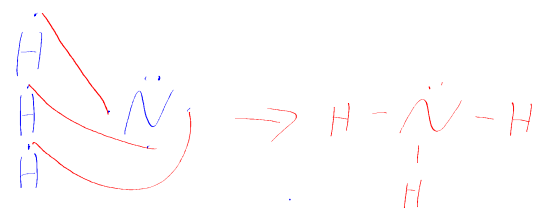
Ex - Molecular Nitrogen (N_2)



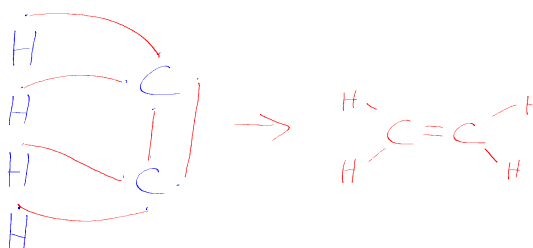
Ex - Water (H_2O)



Ex - Ammonia (NH_3)



Ex Ethene (C_2H_4)



Rules for Naming Binary Covalent Compounds**Rule 1**

The element with the lower group number is written first in the name; the element with the higher group number is written second in the name.

Exception: when the compound contains oxygen and a halogen, the name of the halogen is the first word in the name.

Rule 2

If both elements are in the same group, the element with the higher period number is written first in the name.

Rule 3

The second element in the name is named as if it were an anion, i.e., by adding the suffix -ide to the name of the element.

Rule 4

Greek prefixes are used to indicate the number of atoms of each nonmetal element in the chemical formula for the compound.

Prefix	Number indicated
mono-	1
di-	2
Tri-	3
tetra-	4
penta-	5
hexa-	6
hepta-	7
octa-	8
nona-	9
deca-	10

*chlorine trifluoride**dihydrogen monoxide**carbon tetrachloride*

sulfur hexafluoride
 SF_6



Remember!
Make-up tests tomorrow after school in Rm 232

Write the chemical formula for each of the following molecules:

- | | |
|---|------------------------------|
| (a) nitrogen | (n) sulfur tetrafluoride |
| (b) carbon dioxide | (o) phosphorus pentachloride |
| (c) carbon monoxide | (p) disulfur dichloride |
| (d) nitrogen dioxide | (q) carbon tetrachloride |
| (e) nitrogen monoxide | (r) sulfur trioxide |
| (f) dinitrogen oxide <i>monoxide</i> | (s) sulfur hexafluoride |
| (g) dinitrogen tetroxide | (t) chlorine dioxide |
| (h) sulfur dioxide | (u) dinitrogen pentoxide |
| (i) diiodine pentoxide | (v) phosphorus trichloride |
| (j) silicon tetrafluoride | (w) silicon tetrachloride |
| (k) boron trifluoride | (x) carbon disulfide |
| (l) phosphorus triiodide | (y) phosphorus pentabromide |
| (m) diphosphorus pentoxide | (z) carbon tetrafluoride |

Name the compound indicated by each of the following formulas:

- | | |
|---------------------------------|---------------------------------|
| (a) $\text{SF}_{6(g)}$ | (f) $\text{IF}_{7(g)}$ |
| (b) $\text{N}_2\text{O}_{3(g)}$ | (g) $\text{BF}_{3(g)}$ |
| (c) $\text{NO}_{2(g)}$ | (h) $\text{P}_2\text{S}_{5(s)}$ |
| (d) $\text{PCl}_{3(l)}$ | (i) $\text{P}_2\text{O}_{5(s)}$ |
| (e) $\text{PCl}_{5(s)}$ | |